

04.04 Operations with Complex Numbers

Concept summary and key formulas

Key ideas

Write complex numbers in $a+bi$ form.

Combine real parts with real parts and imaginary parts with imaginary parts.

For division, multiply by the conjugate of the denominator.

Key formulas

$$i^2 = -1$$
$$(a + bi) + (c + di) = (a + c) + (b + d)i$$
$$(a + bi)(a - bi) = a^2 + b^2$$